DO NOW — RECALL

# Logic circuits, adders and flip-flops

Name: _____ Class: _____ Date: _____

**1. Why can a group of four 1s remove two variables?**

**2. Simplify ¬(A ∧ B).**

**3. Calculate 1+1 in binary.**

**4. What is (1 XOR 0) AND 1?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

Two variables change within the group, so only constant variables remain.

### 2. Model answer

¬A ∨ ¬B.

### 3. Model answer

$10_2$.

### 4. Model answer

1. XOR gives 1, and 1 AND 1 is 1.

### Vocabulary — know these terms

half adder — Combinational circuit that adds two one-bit inputs.

full adder — Adds two bits and a carry-in bit.

Sum — Current-column result bit.

CarryOut — Bit passed to the next higher column.

D flip-flop — Clock-triggered circuit that stores one bit.

NEW LEARNING

# Read, annotate and explain

### Find the Sum bit

XOR is 1 when A and B are different; A=1, B=1 gives Sum=0.

### Find the Carry bit

AND is 1 only when both inputs are 1; here Carry=1.

### Extend to full adder and memory

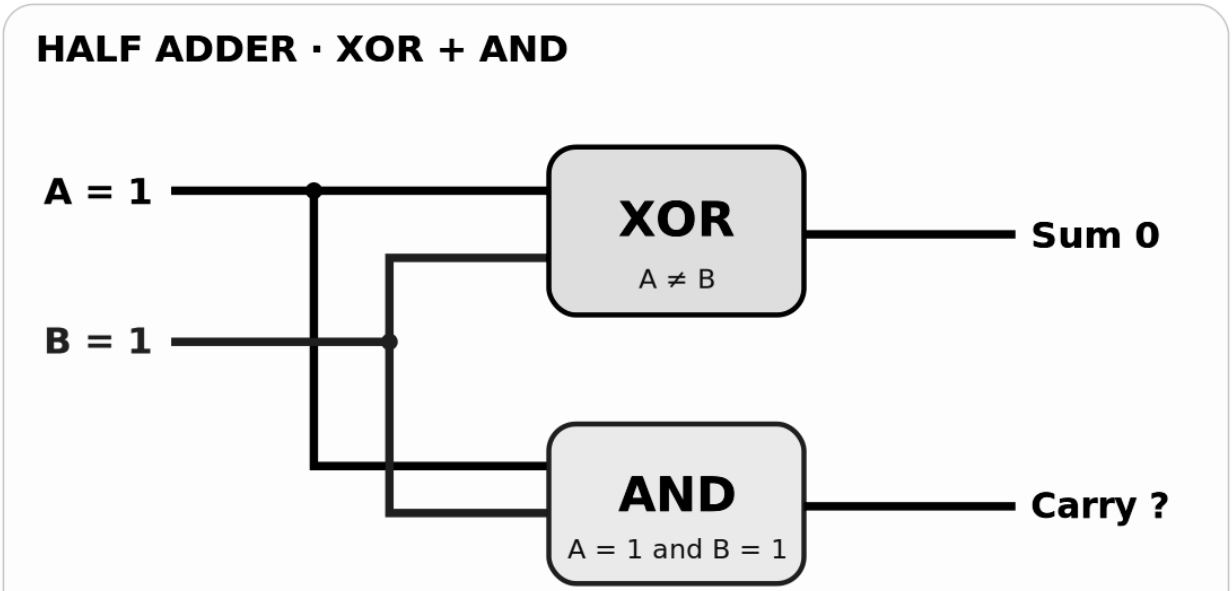
A full adder also includes CarryIn. Unlike combinational logic, a D flip-flop stores a bit until the triggering clock edge.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning

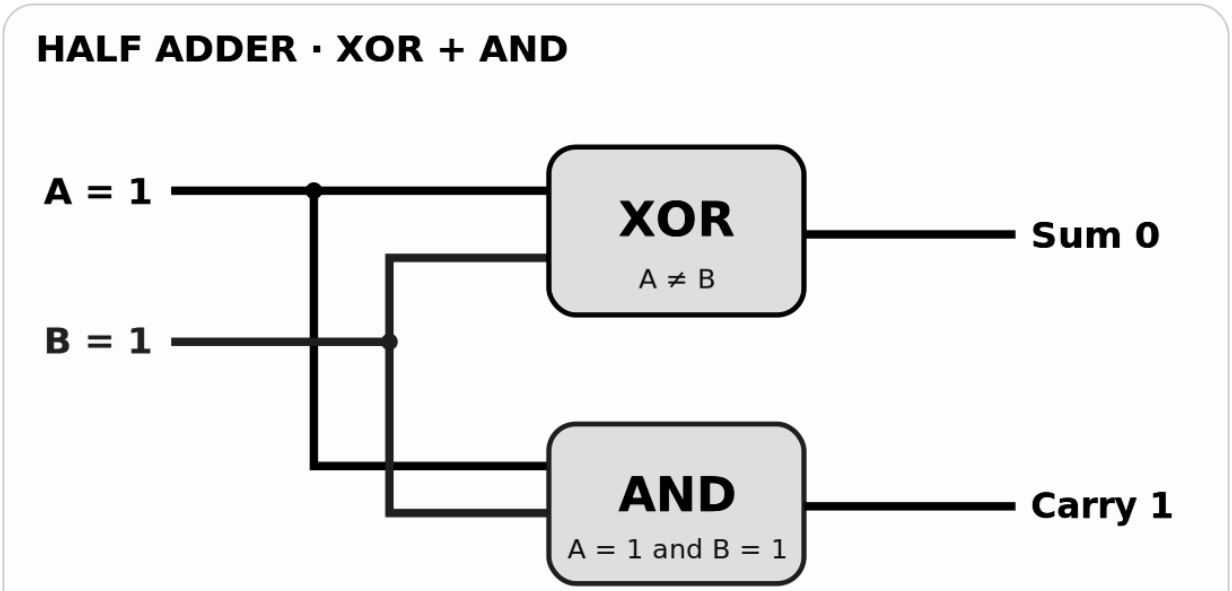



Half-adder logic diagram. A equals 1 and B equals 1. One A wire branches to XOR and AND gates; one B wire also branches to the same gates. Sum 0; Carry 1.

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning

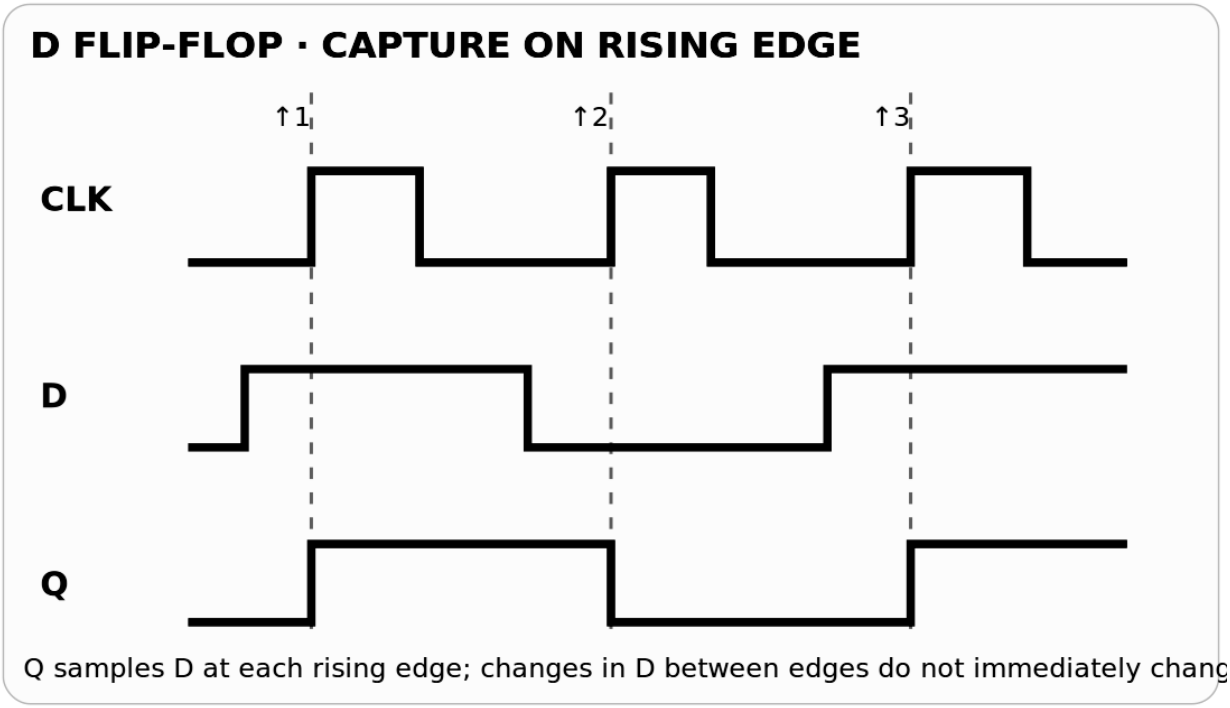



Half-adder logic diagram. A equals 1 and B equals 1. One A wire branches to XOR and AND gates; one B wire also branches to the same gates. Sum 0; Carry 1.

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Rising-edge D flip-flop timing diagram. D is one before the first rising edge, zero before the second and one before the third. Output Q updates only at those three rising edges.

I DO — WORKED EXAMPLE

# Follow the worked method

### A half adder receives A=0, B=1. Calculate Sum and Carry.

### Predict and explain

XOR(0,1)=1, giving Sum=1.

### Check and extend

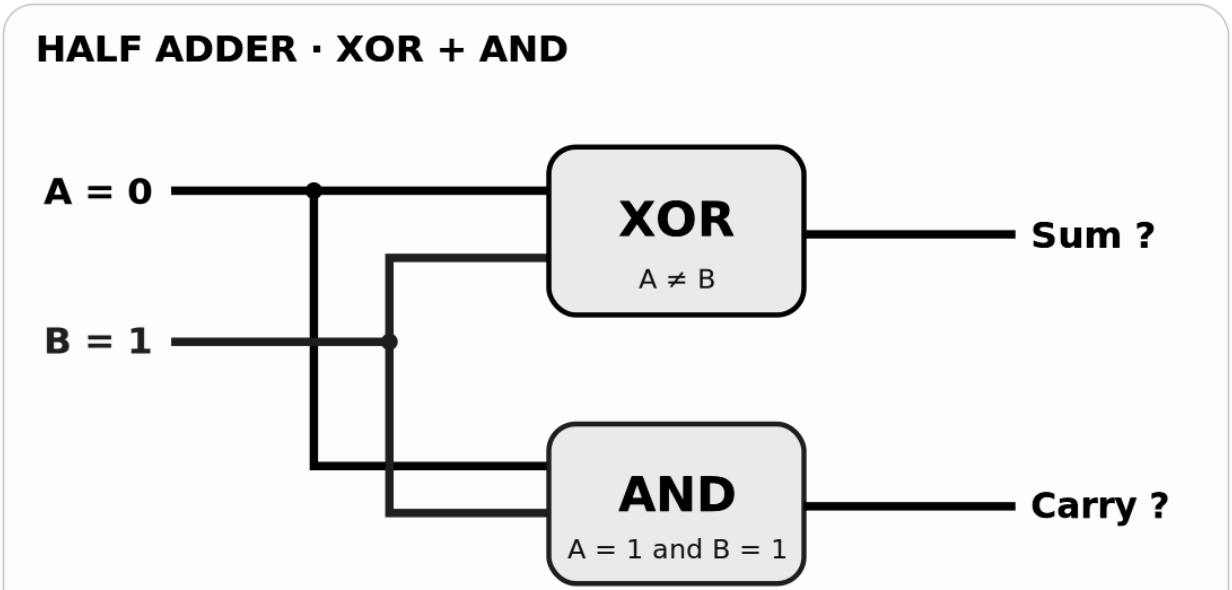
AND(0,1)=0, giving Carry=0.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A half adder receives A=0, B=1. Calculate Sum and Carry.

XOR(0,1)=1, giving Sum=1.



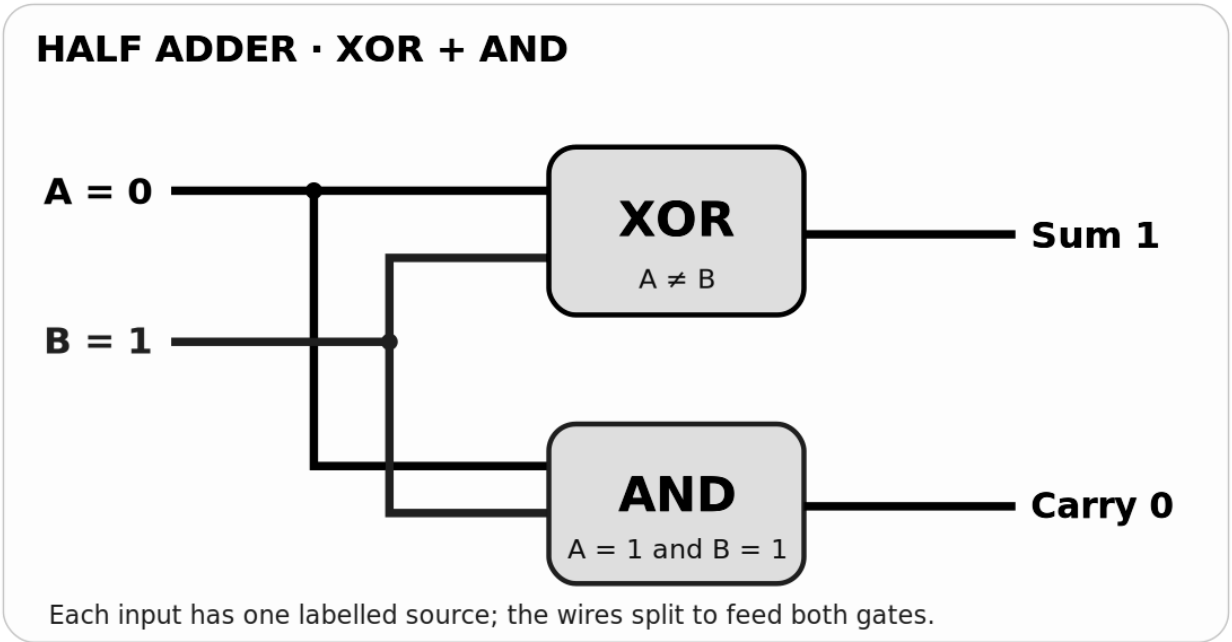

Half-adder logic diagram. A equals 0 and B equals 1. One A wire branches to XOR and AND gates; one B wire also branches to the same gates. Calculate the results.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A half adder receives A=0, B=1. Calculate Sum and Carry.

XOR(0,1)=1, giving Sum=1.




Half-adder logic diagram. A equals 0 and B equals 1. One A wire branches to XOR and AND gates; one B wire also branches to the same gates. Sum 1; Carry 0.

WE DO — GUIDED PRACTICE

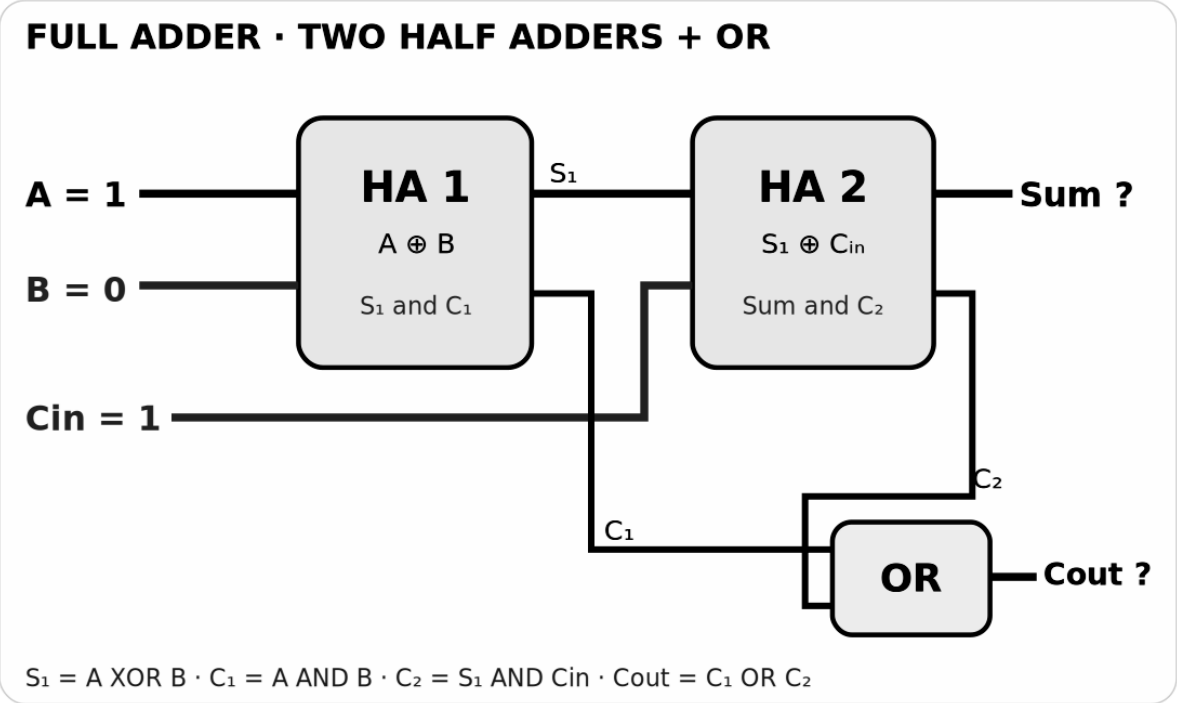
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A full adder receives A=1, B=0, CarryIn=1. Calculate outputs.

The total is decimal 2, binary 10.




Full-adder diagram with three individual input wires A 1, B 0, carry-in 1, two half adders and an OR gate. Predict the outputs.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict and explain The total is decimal 2, binary 10.

DISCUSS — Compare intermediate values and your reasoning. Justify one step using the relevant rule, representation or truth table.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

A half adder receives A=1, B=0. Give Sum and Carry.

### Question 2

For a full adder A=1, B=1 and CarryIn=0, calculate Sum and CarryOut.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

A rising-edge D flip-flop has Q=0 and D=1. D changes to 0 before the next rising clock edge. What value is stored after that edge, and why?

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A D-type flip-flop always changes Q immediately when D changes.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

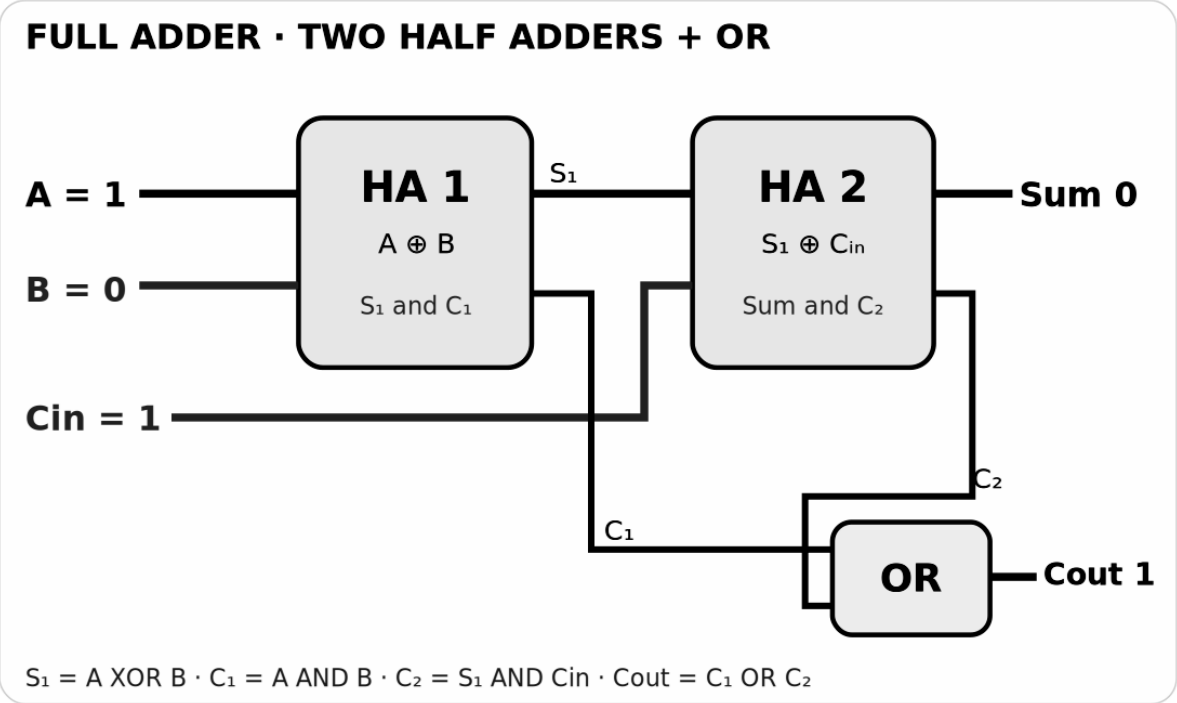
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### A full adder receives A=1, B=0, CarryIn=1. Calculate outputs.

The total is decimal 2, binary 10.




Full-adder diagram with three individual input wires A 1, B 0, carry-in 1, two half adders and an OR gate. Sum 0, carry-out 1.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Check and extend

Sum is the low bit 0; CarryOut is the high bit 1.

IMPROVE: Explain one correction to your own reasoning.